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Exploring Neural Networks for Forward Kinematics of the Robotic Arm with Different Length Configurations: A Comparative Analysis

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***Abstract*—Our work investigates the utilization of Artificial Neural Networks (ANNs) to address the complexities associated with Forward Kinematics (FK) problems within the field of robotics. We undertake an extensive comparative analysis to assess how ANNs perform under different circumstances with robotic arms of varying lengths and impact the overall system's functionality. The training, testing, and validation of ANNs are carried out using MATLAB for a simulated 2-DoF serial robotic arm involving three distinct datasets: fixed step size, random step size, and sinusoidal step size. Three training optimizers, namely Levenberg Marquardt (LM), Bayesian Regularization (BR), and Stochastic Conjugate Gradient (SCG), are considered within the ANN architecture. Based on Mean Square Error (MSE) values, the numerical findings reveal the potential of ANN in estimating forward kinematic solutions of complex robotic manipulators with different arm lengths and reducing computational complexity.**

***Index Terms*—Artificial Neural Networks, Forward kinematics, Robotic manipulators, Random step size, Sinusoidal signal, Training optimizers, Arm lengths**

I. Introduction

Forward kinematics (FK) is a fundamental problem in robotics and computer graphics that involves determining the position and orientation of an end-effector given the joint angles and lengths of robotic arms [1]–[3]. It is essentially the process of predicting where the end-effector will be in space based on the robot's joint configurations. Traditional solutions to the FK problem typically involve using methods from geometry and linear algebra, transformation matrices, and homogeneous coordinate transformations [4], [5].

A well-known method for developing the FK model is the well-known Denavit-Hartenberg (DH) convention. This method uses coordinate frames for joints and the robot's end effector, creating transformation matrices to capture positional and orientational changes between frames. Denavit and Hartenberg's concept involves multiplying these matrices to obtain a global transformation matrix, which describes the journey from the starting point to the final position in the robot system. By inputting joint angles into DH parameters and performing matrix operations, one can determine the end effector's location and orientation relative to the robot's base frame. In [6], authors utilized DH parameters to conduct FK computations for a 6-degree-of-freedom robot featuring a round wrist. The DH measurements were utilized to model the robot's kinematic behavior. In a recent study [7], authors analyzed the FK of a 5-DOF robotic arm, comparing the analytical DH approach and computational approaches in MATLAB.

Another approach proposed for solving FK problems is the product of the exponential (POE) method and the unit dual quaternion method [5]. These methods provide alternative representations for describing the robot's motion. In a comparative study conducted in [8], the POE method and the unit dual quaternion method were evaluated for determining the forward movement of a 7-DoF KUKA LWR robot. The experiment results demonstrated that the unit dual quaternion method is more compact than the POE method.

Authors have employed Artificial Neural Networks (ANNs) and innovative machine learning techniques to tackle FK problems in various types of robotic manipulators [9]–[12]. In [9], an ANN for a five-axis robot arm was used. In [10], authors applied ANN to address singular configurations in parallel manipulators, enhancing their performance. In [11], multilayer ANNs were used for a 7-DOF robotic arm.In [12], a novel approach was introduced, combining classification, ANNs, and particle swarm optimization to address FK in a 4-PUU parallel robot, dividing its workspace into subspaces and using evolutionary ANNs for estimation. Recently, we investigated in [13], [14] the application of ANNs in overcoming the FK using a 2-DoF manipulator arm, using three alternative optimizers and three different datasets. We investigated also in [15] the inverse kinematics problem for this robot.

Despite progress in intelligent techniques for kinematic solutions, we identified gaps in the existing literature, particularly regarding the performance of ANN across diverse robotic arm lengths, dataset variations, and training optimizers

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within ANNs for solving FK in robotics. In this paper, we will conduct a comprehensive assessment using various datasets to address these limitations. There are numerous metrics available for evaluating the performance of the ANN model. In this current study, we exclusively focus on the Mean Squared Error (MSE) as one of the metrics to evaluate the effectiveness of the ANN models in solving the FK of the 2-DoF manipulator robot. The paper is structured as follows: Section II outlines the problem and the proposed method, Section III discusses the used datasets, Section IV presents results for each dataset, Section V contains discussions, and Section VI offers concluding remarks.

II. Problem Formulation and Proposed Method

A. Problem Formulation of FK for 2-DoF Manipulator Robots

By meticulously adhering to the previously outlined procedures and incorporating the DH convention, one can derive the FK model for the 2-DoF manipulator (Fig. 1) as follows.

$$X = l_0 + l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2) \tag{1}$$

$$Y = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2) \tag{2}$$

where θ_1 and θ_2 signify the angles, while l_0, l_1, and l_2 are the lengths associated with distinct segments of the robot arm.

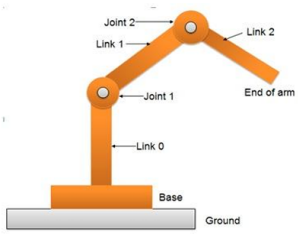

Fig. 1. Joints and Links of the 2-DoF manipulator robot.

Following the above equations, the central idea of this study is first to generate distinct datasets between the joint variables and end-effector positions. Thereafter, a comparative assessment of MSE within the context of ANNs for every generated dataset is carried out. Lesser MSE values are paramount because they directly reflect the accuracy of the model's predictions.

B. Proposed Method

To address this multifaceted challenge comprehensively, a strategic approach has been devised. Firstly, three distinct datasets are thoughtfully selected, each presenting unique characteristics that pose specific challenges to the ANN.

1) *Fixed Step Size Dataset*: This FIX Dataset follows a fixed, uniform pattern, making it relatively predictable.
2) *Random Step Size Dataset*: This RAND Dataset has randomly varying step sizes between data points, testing the ANN's ability to handle noise and adapt to inconsistent data patterns.
3) *Sinusoidal Signal Dataset*: The SIN Dataset assesses the ANN's ability to capture nonlinear/temporal patterns in real-world data with periodic oscillations.

Moreover, to further expand the scope of analysis, two distinct arm-length configurations are introduced:

1) *First Case ($l_0 = 1$, $l_1 = 2$, $l_2 = 3$):* In this configuration, the arm lengths are set to varying values, creating scenarios with different levels of complexity. It challenges the ANN to effectively adapt to varying arm lengths and learn the underlying relationships.
2) *Second Case ($l_0 = 0$, $l_1 = 0.5$, $l_2 = 0.5$):* In contrast, this configuration involves uniform arm lengths, simplifying the mechanical aspects. It focuses on the ANN's ability to handle scenarios where the system's dynamics are more homogeneous.

The primary objective of this multi-faceted experiment is to comprehensively assess the ANN's performance by subjecting it to each dataset within both arm-length configurations.

III. Training with Three Distinct Datasets for Two Arm-Length Value Scenarios

It's crucial to highlight that our parameter choices were made arbitrarily, leading to various potential combinations. While our experiments yielded specific results, it remains the possibility of achieving different values and potentially more optimal models. We methodically adjusted the Hidden Layer count for each model and determined the number of neurons and activation functions. Despite examining multiple configurations, we found minimal variation in the MSE results.

A. ANN with a Fixed Step-Size Dataset

To generate the FIX Dataset for the FK of the 2-DoF robot, the dataset consists of joint angles $(\theta_1,\ \theta_2)$ and the corresponding Cartesian coordinates $(X,\ Y)$ of the robot's end effector. We set the input range for (θ_1, θ_2) from $-\pi$ to π. We iterate through this range with a step size of 0.02. For each combination of joint angles $(\theta_1,\ \theta_2)$, we compute Cartesian coordinates for the given range and step size using FK equations (Eqs. (1) and (2)).

B. ANN with a Random Step-Size Dataset

To generate a RAND Dataset for the FK of the 2-DoF robot, we use the rand function to generate random values for the joint angles. Each data point in the dataset will have two joint angles, θ_1 and θ_2, randomly selected from a uniform distribution. To ensure that the joint angles fall within the desired range, we can scale and shift the random values to obtain joint angles ranging from $-\pi$ to π.

C. ANN with a Sinusoidal Signal-Based Dataset

During the training process of the ANN, we integrated the input-output datasets, ensuring that the network received both the input data and their corresponding expected outputs. To compile the ANN model or algorithm, we carefully incorporated the necessary hyperparameters shown in Table I.

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TABLE I
PARAMETERS USED FOR GENERATING THE SIN DATASET.

Frequency	1.5	10
Phase	0	$\pi/4$
Amplitude	π	π
Angle	$[-\pi\ \pi\ ]$	$[-\pi\ \pi\ ]$
Number of samples	1000	1000

IV. ANN RESULTS FOR FK PROBLEMS

In our experiments with different datasets and training optimizers, we examined the performance of ANNs with varying numbers of Hidden Layers, specifically comparing networks with one Hidden Layer to five Hidden Layers. Fig. 2 shows an example of an ANN model with three Hidden Layers. Our objective was to assess the impact of increasing the number of Hidden Layers on the network's capacity to learn and represent complex patterns. For every dataset, three training optimizers are considered within the ANN model: Levenberg Marquardt (LM) [16], Bayesian Regularization (BR) [17], [18], and Stochastic Conjugate Gradient (SCG) [19].

Fig. 2. ANN architecture with three Hidden Layers.

A. Results of the First Case

1) Training of Three Datasets with LM Algorithm: As shown in Fig. 3, for the FIX Dataset, the best MSE achieved was 9.4883e-5 with 3 Hidden Layers. On the other hand, when training the ANN with RAND Dataset, the best MSE obtained was 14.873e-5 with 4 Hidden Layers. Lastly, training the ANN with the SIN Dataset resulted in the best MSE of 86.455e-5 at 1000 epochs. Similarly to the RAND Dataset scenario, this model also employed 4 Hidden Layers. More details on the number of neurons in the Hidden Layers and respective MSE values are shown in Table II.

2) Training of Three Datasets with BR Algorithm: Fig. 4 displays the best training results of an ANN with the BR algorithm on three different datasets. The BR algorithm was used to prevent overfitting and enhance generalization. The best MSE achieved for FIX Dataset was 9.1609e-05 at epoch 63, while the ANN had 3 Hidden Layers with varying neuron counts. For RAND Dataset, the best MSE was 0.00012283 at epoch 135, using 4 Hidden Layers. For SIN Dataset, the best MSE was 0.00010382 at epoch 60, with 3 Hidden Layers. Table III contains additional results.

3) Training of Three Datasets with SCG Algorithm: Fig. 5 displays the best training results of an ANN with the SCG algorithm on three different datasets. When trained with RAND Dataset, the ANN achieved an MSE of 0.45943 at epoch 684 with a single Hidden Layer of 10 neurons. When trained with SIN Dataset, the ANN achieved an MSE of 0.16541 at epoch 1000 with a model architecture consisting of 4 Hidden Layers with varying numbers of neurons. Similarly, training the ANN with SIN Dataset yielded the best MSE of 0.16541 at epoch 1000 with 4 Hidden Layers. The detailed results can be found in Table IV.

B. Results of the Second Case

1) Training of Three Datasets with LM Algorithm: Fig. 6 displays the training results of an ANN with the LM algorithm on three different datasets. The best performance with very low MSE (9.8707e-5) on a FIX Dataset is observed for 4 Hidden Layers at 248 epochs. A 3-layered architecture at 1000 epochs achieved low MSE (9.3575e-5) on a RAND Dataset. For the SIN Dataset, a 4-layered architecture at 345 epochs also had low MSE (9.1325e-5). This highlights the significant influence of Hidden Layer architecture on ANN performance across different datasets. The detailed information on Hidden Layer configurations is tabulated in Table V.

2) Training of Three Datasets with BR Algorithm: Fig. 7 illustrates the best-performing results of an ANN with the BR algorithm on three different datasets. In the case of the FIX Dataset, the least MSE was achieved at 9.7948e-05, utilizing a network architecture with four Hidden Layers at 129 epochs. Similarly, for the RAND Dataset, the lowest MSE was 8.5087e-05, also attained with a four-layered network at 38 epochs. On the other hand, the SIN Dataset exhibited its best performance with an MSE of 0.00010209 and an epoch count of 704, using a simpler network structure consisting of just one Hidden Layer with 10 neurons. The detailed analysis with the number of neurons in each layer is presented in Table VI.

3) Training of Three Datasets with SCG Algorithm: The best-performing results (lowest MSE), of an ANN with the SCG algorithm on three different datasets are shown in Fig. 8. The FIX Dataset worked best (MSE of 0.011521) with one Hidden Layer with 10 neurons. The RAND Dataset required more complexity, with two Hidden Layers achieving the lowest MSE of 0.022738. The SIN Dataset performed similarly to RAND Dataset, achieving MSE of 0.036255 from two Hidden Layers. Detailed configurations and MSE values are in Table VII.

V. DISCUSSIONS

We analyze the impact of datasets and optimization methods on MSE in ANNs using three datasets. In the first case (arm lengths $l_0 = 1$, $l_1 = 2$, $l_2 = 3$), FIX Dataset worked best for LM and BR algorithms, achieving low MSE. SCG excelled with a SIN Dataset and 4 Hidden Layers. In the second case (arm lengths $l_0 = 0$, $l_1 = 0.5$, $l_2 = 0.5$), LM performed optimally with the sin dataset and 4 Hidden Layers, BR with RAND Dataset and 4 Hidden Layers, and SCG with the FIX Dataset and one Hidden Layer. Simplifying the problem's complexity led to better error minimization, emphasizing the importance of tailored optimization in ANNs.

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TABLE II
RESULTS OF TRAINING WITH LM ALGORITHM FOR DIFFERENT DATASETS AND FOR THE FIRST CASE.

Hidden Layers (Neurons)	FIX Dataset	RAND Dataset	SIN Dataset
1 Hidden Layer (10)	MSE: 0.0015146 Epoch: 1000	MSE: 0.00087046 Epoch: 365	MSE: 0.0057597 Epoch: 319
2 Hidden Layers (10,2)	MSE: 0.0030593 Epoch: 1000	MSE: 0.00094045 Epoch: 1000	MSE: 0.00086455 Epoch: 1000
3 Hidden Layers (10,20,2)	MSE: 9.4883e-05 Epoch: 99	MSE: 0.019793 Epoch: 112	MSE: 4.5768 Epoch: 9
4 Hidden Layers (10,20,5,2)	MSE: 9.8986e-05 Epoch: 141	MSE: 0.00041854 Epoch: 254	MSE: 0.00014873 Epoch: 247
5 Hidden Layers (10,20,10,5,2)	MSE: 9.8327e-05 Epoch: 158	MSE: 0.0021619 Epoch: 270	MSE: 3.4841 Epoch: 8

TABLE III
RESULTS OF TRAINING WITH BR ALGORITHM FOR DIFFERENT DATASETS AND FOR THE FIRST CASE.

Hidden Layers (Neurons)	FIX Dataset	RAND Dataset	SIN Dataset
1 Hidden Layer (10)	MSE: 0.00029686 Epoch: 1000	MSE: 0.00020188 Epoch: 765	MSE: 0.0024345 Epoch: 81
2 Hidden Layers (10,2)	MSE: 0.00020449 Epoch: 1000	MSE: 0.00023001 Epoch: 1000	MSE: 0.0025418 Epoch: 185
3 Hidden Layers (10,20,2)	MSE: 9.1609e-05 Epoch: 63	MSE: 0.00013403 Epoch: 61	MSE: 0.00010382 Epoch: 60
4 Hidden Layers (10,20,5,2)	MSE: 0.00010186 Epoch: 177	MSE: 0.00012283 Epoch: 135	MSE: 0.00013139 Epoch: 75
5 Hidden Layers (10,20,10,5,2)	MSE: 0.00010104 Epoch: 110	MSE: 0.00013012 Epoch: 117	MSE: 0.00010982 Epoch: 172

TABLE IV
RESULTS OF TRAINING WITH SCG ALGORITHM FOR DIFFERENT DATASETS AND FOR THE FIRST CASE.

Hidden Layers (Neurons)	FIX Dataset	RAND Dataset	SIN Dataset
1 Hidden Layer (10)	MSE: 0.17430 Epoch: 1000	MSE: 0.45943 Epoch: 684	MSE: 1.12660 Epoch: 373
2 Hidden Layers (10,2)	MSE: 0.63463 Epoch: 1000	MSE: 0.63920 Epoch: 951	MSE: 0.63969 Epoch: 734
3 Hidden Layers (10,20,2)	MSE: 0.26303 Epoch: 1000	MSE: 0.90189 Epoch: 373	MSE: 0.18672 Epoch: 1000
4 Hidden Layers (10,20,5,2)	MSE: 0.18089 Epoch: 1000	MSE: 1.413 Epoch: 570	MSE: 0.16541 Epoch: 1000
5 Hidden Layers (10,20,10,5,2)	MSE: 0.90833 Epoch: 999	MSE: 1.4275 Epoch: 547	MSE: 2.22790 Epoch: 443

TABLE V
RESULTS OF TRAINING WITH THE LM ALGORITHM FOR DIFFERENT DATASETS AND FOR THE SECOND CASE.

Hidden Layers (Neurons)	FIX Dataset	RAND Dataset	SIN Dataset
1 Hidden Layer (10)	MSE: 9.8951e-05 Epoch: 959	MSE: 0.0029241 Epoch: 333	MSE: 0.00010245 Epoch: 252
2 Hidden Layers (10,2)	MSE: 0.00011872 Epoch: 1000	MSE: 0.00010455 Epoch: 939	MSE: 0.01079 Epoch: 999
3 Hidden Layers (10,20,2)	MSE: 0.00015033 Epoch: 708	MSE: 9.3575e-05 Epoch: 1000	MSE: 9.8624e-05 Epoch: 391
4 Hidden Layers (10,20,5,2)	MSE: 9.8707e-05 Epoch: 248	MSE: 0.0036766 Epoch: 219	MSE: 9.1325e-05 Epoch: 345
5 Hidden Layers (10,20,10,5,2)	MSE: 0.0001006 Epoch: 151	MSE: 0.00012044 Epoch: 162	MSE: 0.026683 Epoch: 160

TABLE VI
RESULTS OF TRAINING WITH THE BR ALGORITHM FOR DIFFERENT DATASETS AND FOR THE SECOND CASE.

Hidden Layers (Neurons)	FIX Dataset	RAND Dataset	SIN Dataset
1 Hidden Layer (10)	MSE: 0.00010206 Epoch: 208	MSE: 0.00010731 Epoch: 156	MSE: 0.00010209 Epoch: 704
2 Hidden Layers (10,2)	MSE: 0.00010002 Epoch: 443	MSE: 0.00014205 Epoch: 29	MSE: 0.00010358 Epoch: 145
3 Hidden Layers (10,20,2)	MSE: 9.9332e-05 Epoch: 714	MSE: 0.00011406 Epoch: 97	MSE: 0.00010449 Epoch: 772
4 Hidden Layers (10,20,5,2)	MSE: 0.00010216 Epoch: 134	MSE: 8.5087e-05 Epoch: 38	MSE: 0.00010794 Epoch: 43
5 Hidden Layers (10,20,10,5,2)	MSE: 9.7948e-05 Epoch: 129	MSE: 0.00011042 Epoch: 53	MSE: 0.0001075 Epoch: 63

TABLE VII
RESULTS OF TRAINING WITH THE SCG ALGORITHM FOR DIFFERENT DATASETS AND FOR THE SECOND CASE.

Hidden Layers (Neurons)	FIX Dataset	RAND Dataset	SIN Dataset
1 Hidden Layer (10)	MSE: 0.011521 Epoch: 1000	MSE: 0.1228 Epoch: 949	MSE: 0.051858 Epoch: 286
2 Hidden Layers (10,2)	MSE: 0.026335 Epoch: 1000	MSE: 0.022738 Epoch: 966	MSE: 0.036255 Epoch: 553
3 Hidden Layers (10,20,2)	MSE: 0.045194 Epoch: 1000	MSE: 0.090759 Epoch: 299	MSE: 0.087149 Epoch: 267
4 Hidden Layers (10,20,5,2)	MSE: 0.034478 Epoch: 1000	MSE: 0.10814 Epoch: 244	MSE: 0.14413 Epoch: 96
5 Hidden Layers (10,20,10,5,2)	MSE: 0.057815 Epoch: 1000	MSE: 0.060233 Epoch: 583	MSE: 0.08326 Epoch: 428

VI. CONCLUSION AND FUTURE WORKS

In conclusion, our study reveals that dataset complexity and optimization methods significantly impact MSE in ANNs. Choosing the right dataset and algorithm is crucial for network performance. Tailoring optimization strategies to specific problems is essential.

Future research should apply these optimized ANNs in real-world scenarios like testing on robot SCARA arms to advance robotics and automation. Moreover, the idea is to incorporate revolutionary intelligent algorithms to optimize and hence find the optimal ANN architecture.

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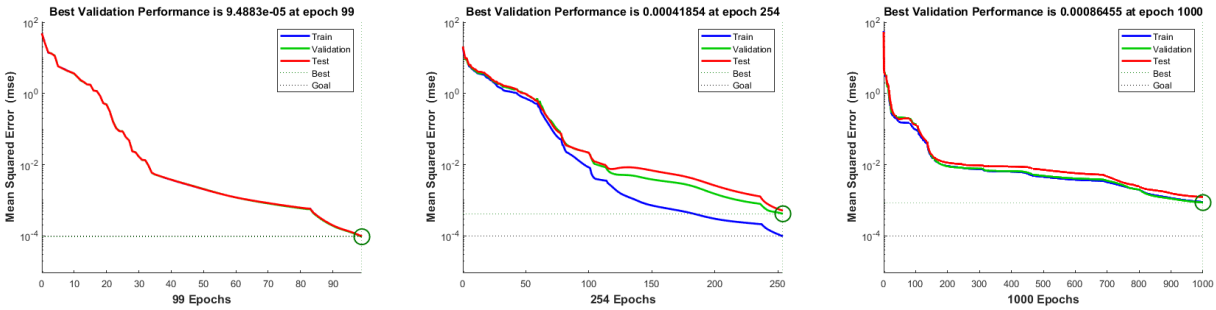
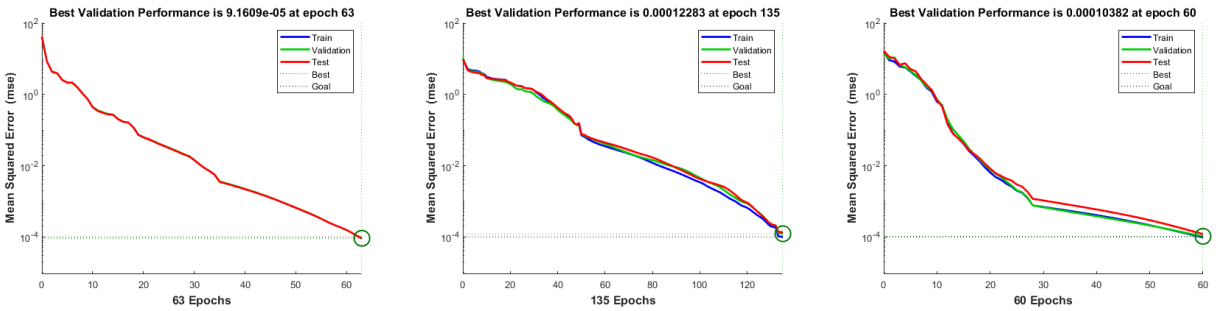


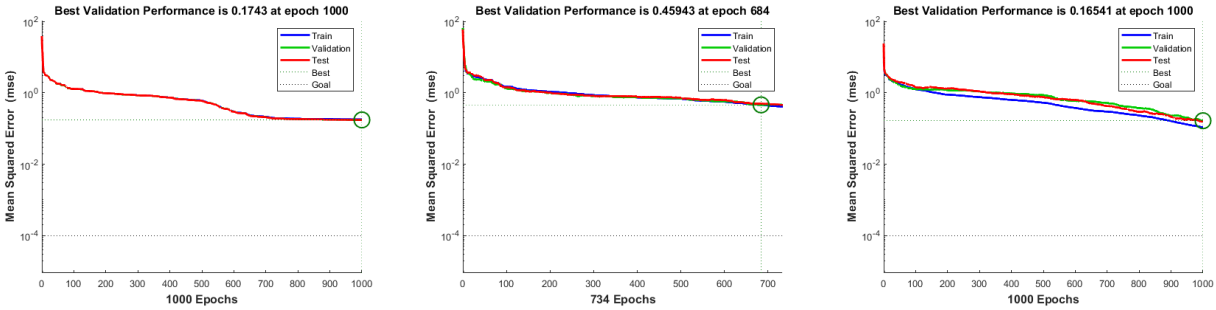
(a) Lowest MSE for FIX Dataset (b) Lowest MSE for RAND Dataset (c) Lowest MSE for SIN Dataset

Fig. 3. Best ANN's performances with LM algorithm for the first case.



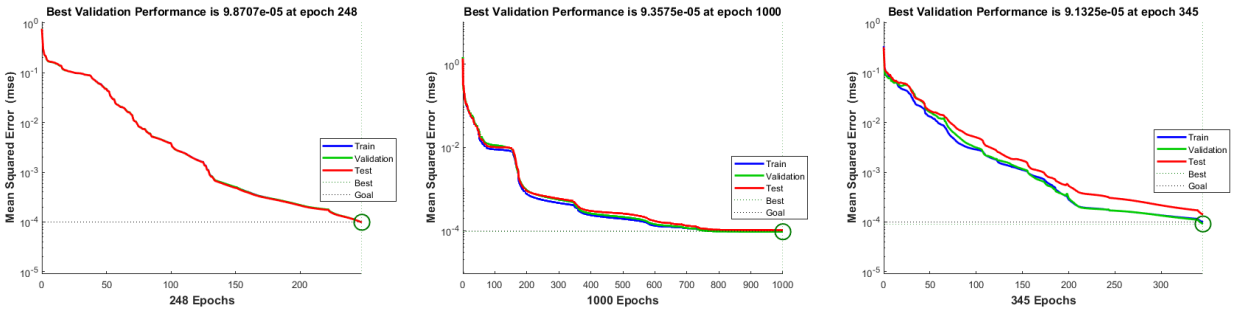
(a) Lowest MSE for FIX Dataset (b) Lowest MSE for RAND Dataset (c) Lowest MSE for SIN Dataset

Fig. 4. Best ANN's performances with BR algorithm for the first case.



(a) Lowest MSE for FIX Dataset (b) Lowest MSE for RAND Dataset (c) Lowest MSE for SIN Dataset

Fig. 5. Best ANN's performances with SCG algorithm with the first case.



(a) Lowest MSE for FIX Dataset (b) Lowest MSE for RAND Dataset (c) Lowest MSE for SIN Dataset

Fig. 6. Best ANN's performances with LM algorithm for the second case.

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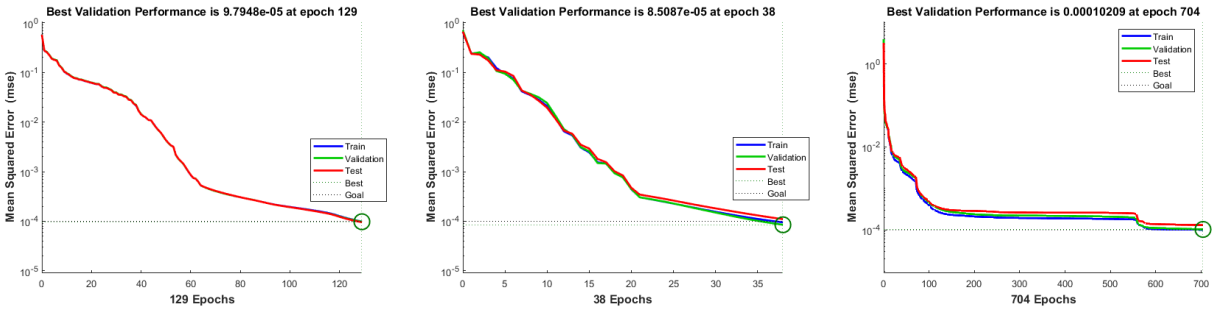
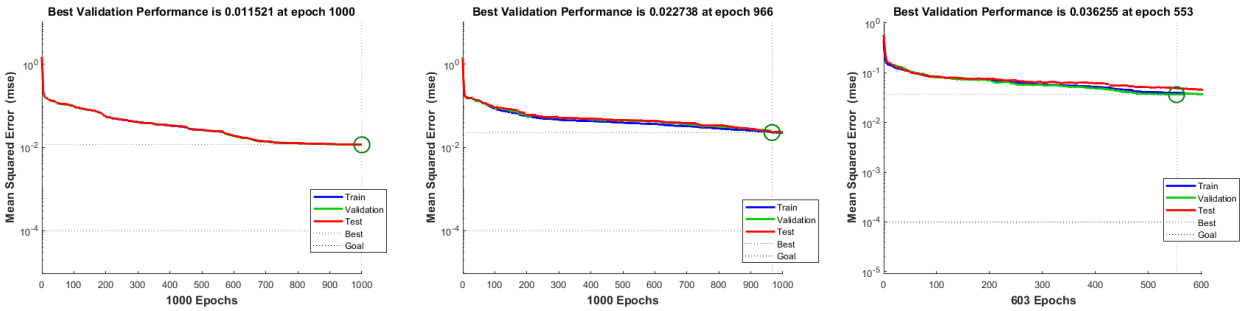


(a) Lowest MSE for FIX Dataset (b) Lowest MSE for RAND Dataset (c) Lowest MSE for SIN Dataset

Fig. 7. Best ANN's performances with BR algorithm for the second case.



(a) Lowest MSE for FIX Dataset (b) Lowest MSE for RAND Dataset (c) Lowest MSE for SIN Dataset

Fig. 8. Best ANN's performances with SCG algorithm for the second case.

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